

White test N°6 - RHCSA 8

Initialization : Systems set up

System1 :

Hostname – system1.eight.example.com

IP – 192.168.55.150/24

Dns – 192.168.5.1

GW – 192.168.5.1

`Hostnamectl --set--hostname system1.eight.example.com`

`Nmcli connection add ifname ens160 type ethernet ipv4.addresses 192.168.55.150/24 ipv4.dns 192.168.5.1 ipv4.gateway 192.168.5.1`

System2 :

Hostname – system2.eight.example.com

IP – 192.168.55.151/24

Dns – 192.168.5.1

GW – 192.168.5.1

`Hostnamectl --set--hostname system2.eight.example.com`

`Nmcli connection add ifname ens160 type ethernet ipv4.addresses 192.168.55.151/24 ipv4.dns 192.168.5.1 ipv4.gateway 192.168.5.1`

Questions

System1 :

- 1- The system administrator has just resigned from your organisation abruptly, as the new administrator reset the root password of the system1 server to « password123 »
- 2- create a backup file named « /root/backup.tar.bz2 ». the backup file should contain the content of « /usr/local » and should be zipped with bzip2 compression format.
`su root -c "tar -cjf /root/backup.tar.bz2 /usr/local"`
- 3- Ensure selinux is in enforcing mode. If it is not, change selinux to enforcing mode.

`Getenforce`

`Setenforce 1`

- 4- As the student user (create a user « student »), create a detached apache http web server container with the name, (site1) and with the tag that has the lowest version1 from [rhel8/httpd-24 image](#). Use the [registry.redhat.io.registry](#) you can use the username administrator and the password admin123.

- 5- Create and mount the ~/storage/html/ directory as a persistent storage to the container as /var/www/. The content of ~/storage/html should be extracted from /tmp/image/containers.tgz

Also, port 8080 on the container should be mapped to port 8080 on the host. Declare the environment variables, httpd_user and httpd_password and use admin as their values.

- 6- Configure the container as a service using systemd and make the web server/container persistent across reboot.
- 7- Install a FTP server, and request to anonymous download from /var/ftp/pub catalog. (it needs you to configure yum direct to the already existing file server.)
- 8- Upgrade the kernel to kernel-5.15.13-193.14.3.el8_2.x86_64, and configure the system to Start with the default kernel. Also, keep the old kernel available.
Verify if the appropriate module is available : `Dnf list kernel --showduplicates`
`Dnf upgrade-to kernel-5.15.13-193.14.3.el8_2`
- 9- Configure a task : plan to run echo hello command at 7:13 every first day of each month.
`Crontab -e`
`13 7 */1 * * echo 'Hello'`
- 10- Find the files owned by the user student, and copy it to catalog : /opt/dir
`find / -user "student" -exec cp -r {} /opt/dir/catalogue \; 2>/dev/null`
`su root -c "locate -w 'student' 1>/opt/dir/catalogue 2>/dev/null" => works`
- 11- Find the rows that contain nologin from file /etc/passwd, and write it to the file /tmp/testfile.
`Grep 'nologin' /etc/passwd`
- 12- Configure the system synchronous as 192.168.55.151 (system2).

- 13- Add user1, user2 and user3. The additional group of the two users : user2, user3 is the admin group. Password : redhat

```
For i in user{1,2,3}
Usermod -aG admin user{2,3}
Passwd user1
Passwd user2
Passwd user3
```

- 14- Add admin group and set gid=6000

```
Groupadd admin -g 6000
```

- 15- Create a shared directory /home/admins, make it has the following characteristics :

/home/admins belongs to group administrator

This directory can read and written by members of group adminuser.

Any files created in /home/admins must has adminuser as owner group.

```
Mkdir /home/admins
```

```
Chown :administrator /home/admins
Setfacl -R -m g :adminuser :rw- /home/admins
Chmod g+s /home/admins
```

- 16- Copy /etc/fstab to /var/tmp/fstab. Owner of the file /var/tmp/fstab is root, belongs to group root, and this file can not be executed by any user, the user1 could read, write and modify it, while user2 without any permission.

```
su root -c "cp /etc/fstab /var/tmp/fstab"
chmod 700 /var/tmp/fstab
setfacl -R -m u:user1 :rwx /var/tmp/fstab
setfacl -R -m u:user2:--- /var/tmp/fstab
```

- 17- Write a replace.sh script which receives as parameters a file name Fich1 and a directory dir. The file file1, the directory dir and the script replace.sh are in the current directory.

- After the execution of the script, any file in rep must have its contents replaced by that of file1.
- This script must also calculate the number of elements contained in rep. (x files and y directories).

```
#!/bin/bash
workd=`pwd`
files=$(ls -l | cut -f1 -d" " | grep "^-" | wc -l)
until [[ $f != "" && $d != "" ]]
do
    read -p "Enter your file name : " f
    read -p "Enter your directory name : " d
    if [ ! -f $f ]; then
        echo "$f not found"
    elif [ ! -d $d ]; then
        echo "$d not found"
    fi
done
lists=$(ls -l | awk '{print $9}' > result)
#for i in $(cat result) ;do($(cat f > i)) 2>/dev/null;done {First try}
for i in $(cat .result);do (cat $f > $i) 2>/dev/null;done

echo "Nombres des fichiers dans $workd est $(ls -l | cut -f1 -d" " | grep "^-" | wc -l)"

echo "Nombres des repertoires dans $workd est $(ls -l | cut -f1 -d- | grep "d" | wc -l)"
```

- 18- Open the ip_forward, and take effect permanently. (rpc-bind ?)

- 19- The firewall must be open

- 20- Configure your NFS services. Share the directory by the NFS shared services.

System2 :

- 21- Create a 512M partition (/dev/sdb of 2GB), make it as ext4 file system, mounted automatically under /mnt/data1 and which take effect automatically at boot-start

```
Mkdir /mnt/data1
Parted
    Mklabel msdos
    Mkpart primary
    Ext4
    Start 1MiB
    End 513MiB
    Q

blkid /dev/nvme0n4p1

echo 'UUID= 'xxxxxxxx' /mnt/data1 ext4 defaults 0 0' | sudo tee -a /etc/fstab

mount -a
```

- 22- Make a swap partition having 100MB (/dev/sdb of 2GB). Make automatically usable at system boot time.

Warning: You requested a partition from 1049kB to 106MB (sectors 2048..206847).

The closest location we can manage is 1048kB to 1048kB (sectors 2047..2047).

Is this still acceptable to you? => question

```
Fdisk /dev/nvme0n4

N

2

Primary

+100M

t=> 82 linux lvm

w

mkswp /dev/nvme0n4p2

swapon /dev/nvme0n4p2

sudo blkid /dev/nvme0n4p2

vim /etc/fstab

UUID= XXXXX none swap defaults 0 0

Mount -a
```

- 23- Create a volume group vg (dev/sdc of 3GB), and set 16M as a extends. And divided a volume group containing 50 extends on 2 logical volumes lvo1 and lvo2, make it as ext4 file system, and mounted automatically under /mnt/lv1 and /mnt/lvo2. And the size of the floating range should set between 380M and 400M.
- 24- Adjust the size of the logical volume lvo1 , its file system size should be 890M. Make sure that the content of this system is complete.
- 25- (/dev/sdc of 10GB) : create a VDO volume with the name (class1_vdo) and a logical size of 30GB. Format the VDO volume with the xfs filesystem, mount it on /class1_mnt and make it persistent across reboot.